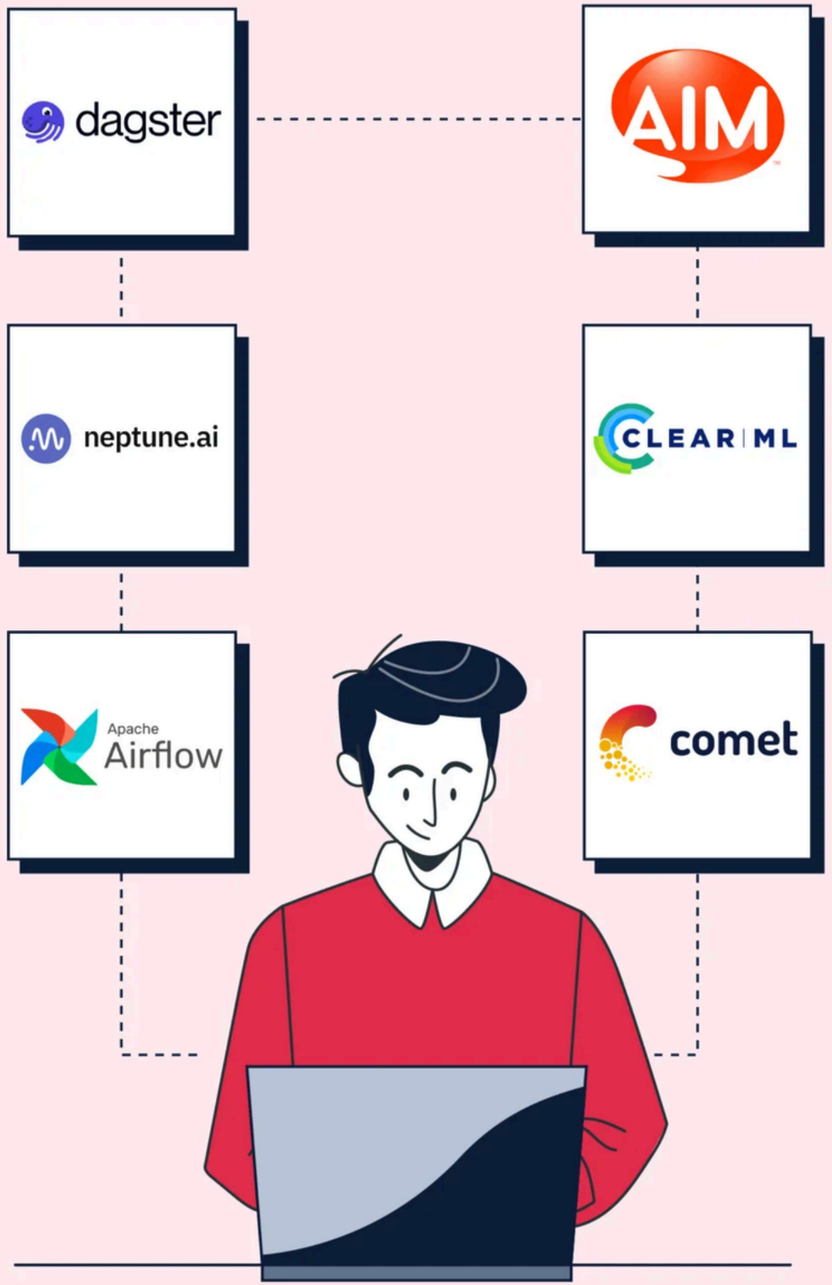
SriRam's

MLOps Tools



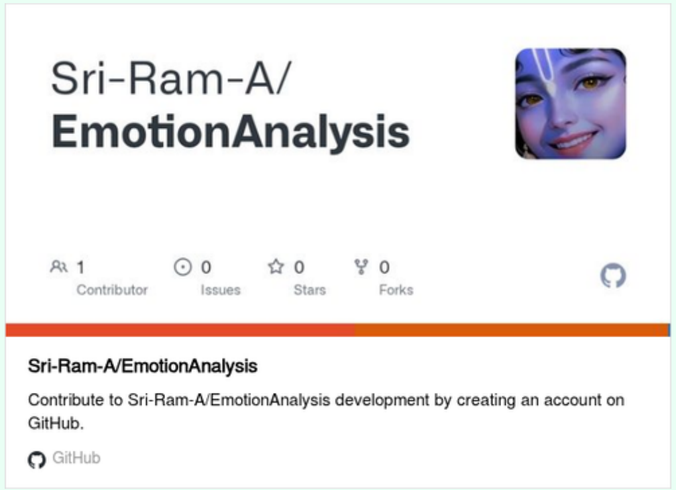
What is Sentiment Analysis ?

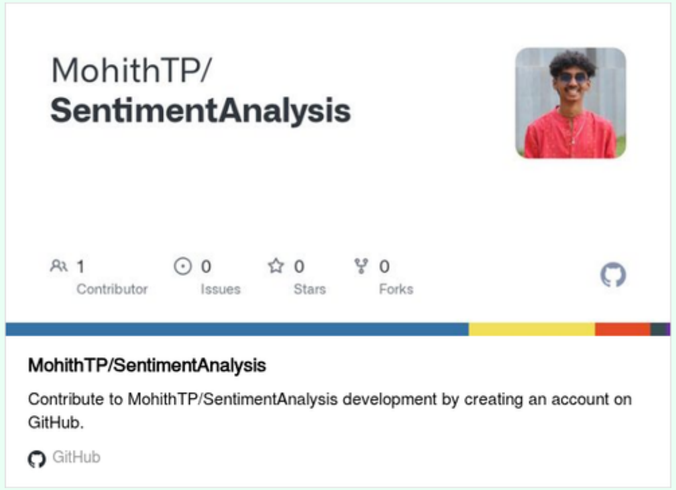
What is Sentiment Analysis?

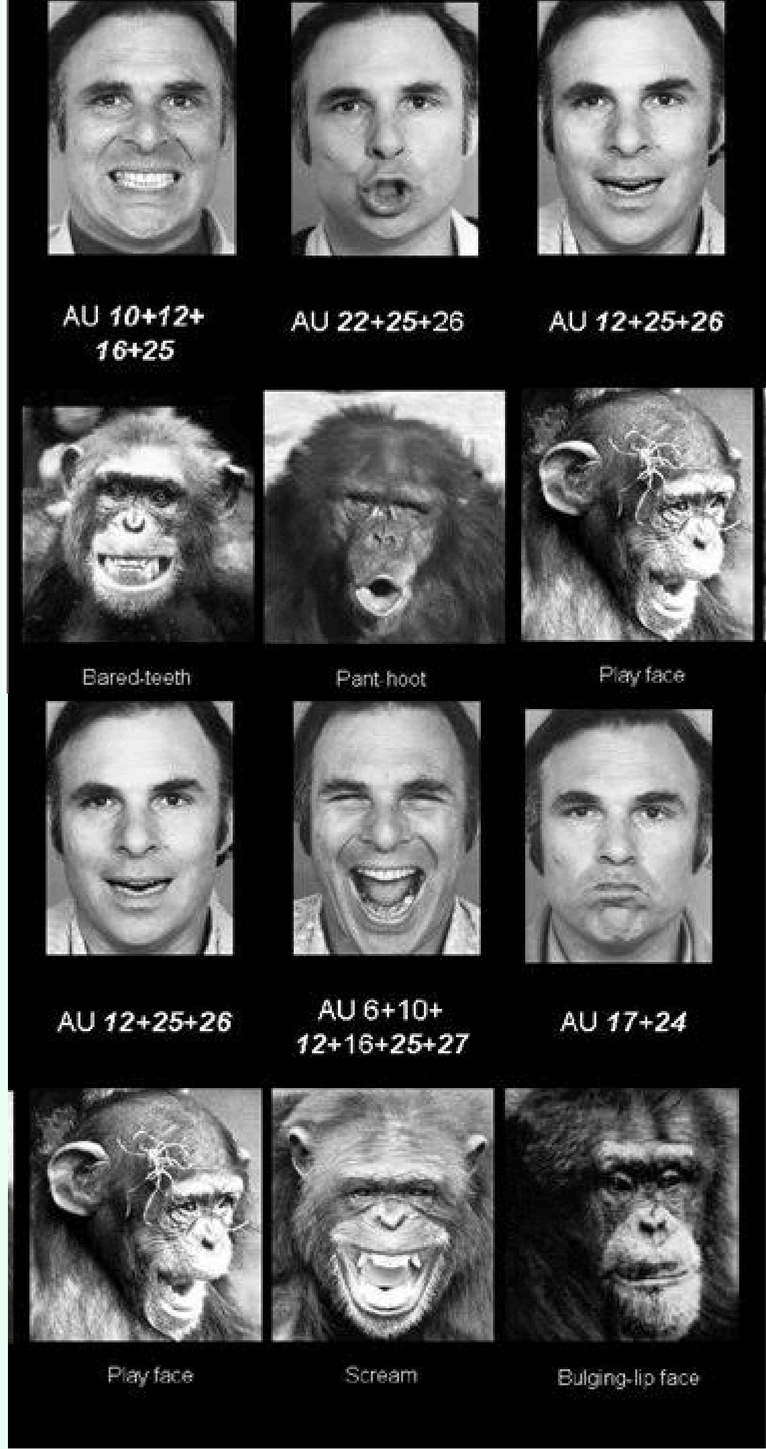
Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify and classify human emotions expressed in text data.

It goes beyond simple sentiment (positive/negative) and detects specific emotions such as:

- Joy
- Sadness
- Anger
- Fear
- Neutral





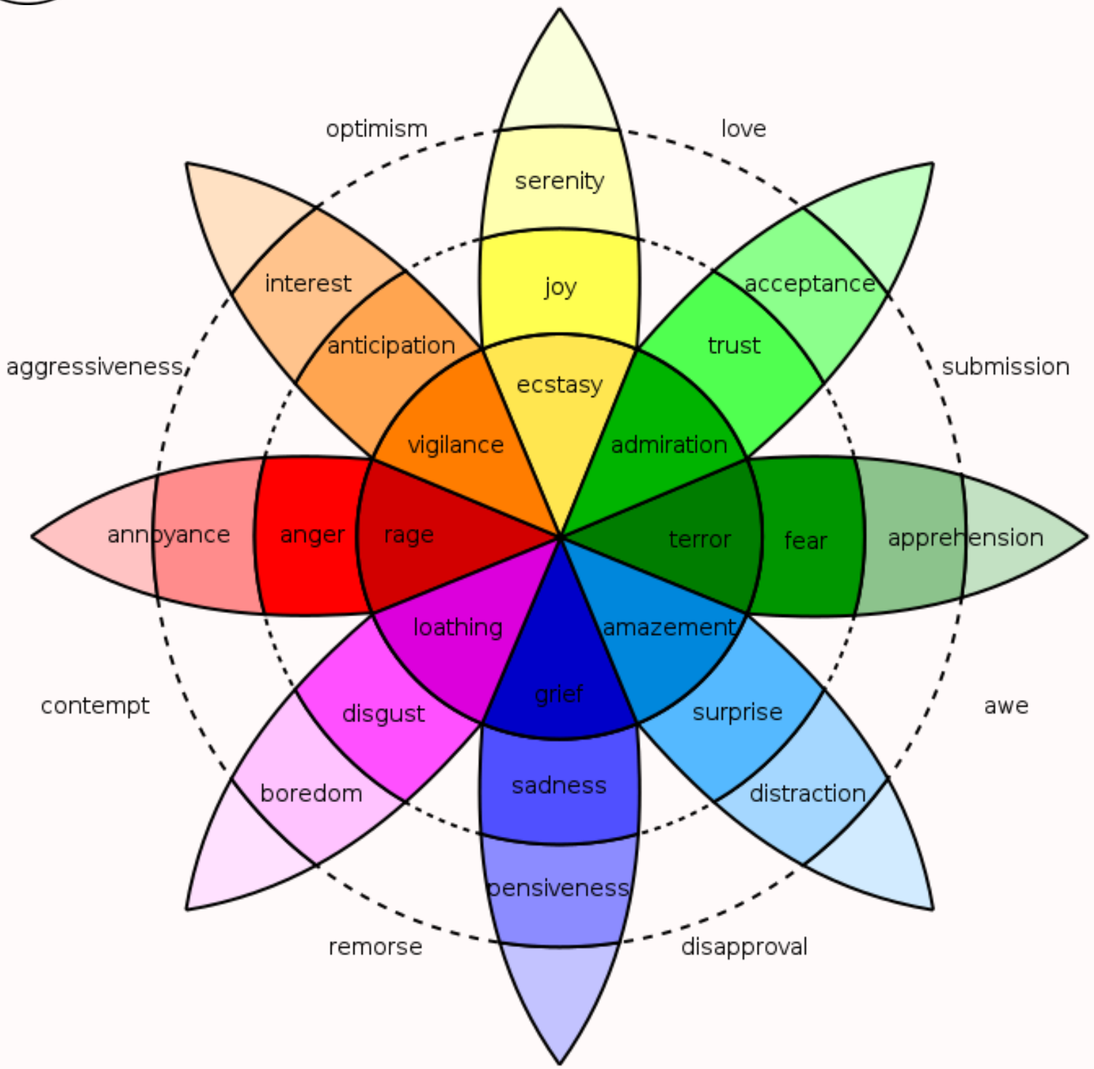


Team No :11

Machine Learning Operations (MLOps)

SENTIMENT ANALYSIS

SriRam A | Srujan | Sumith | TP Mohith



Motivation – Real-World Example (Social Media & Chatbots)

Every day, millions of users express emotions through:

- Social media posts
- Customer reviews
- Chat messages

Manual emotion understanding is:

- Time-consuming
- Not scalable
- Subjective

PROBLEM STATEMENT

Large amounts of text from social media, customer feedback, and online communication contain important emotional insights.

Manually identifcation from this unstructured data is slow, subjective, and not scalable.

This project develops an automated emotion analysis system using machine learning and natural language processing to accurately classify emotions from text, while applying MLOps practices to deploy, manage, and monitor the model efficiently.

OBJECTIVES

1. To collect and preprocess text-data for classification using NLP techniques.
2. To build and evaluate machine learning and deep learning models for emotion detection.
3. To optimize model performance using hyperparameter tuning techniques.
4. To deploy the trained emotion analysis model as a scalable web service.
5. To implement MLOps practices for automation, version control, monitoring, and reproducibility.

DATASET DESCRIPTION

We train on three complementary datasets.

- The Kaggle Twitter Sentiment dataset (1.6M tweets, 2009) provides scale and realistic social media language; we use the text & map binary sentiment to our 7 emotion classes.

(https://www.kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis)

- GoEmotions adds 58K Reddit comments with high-quality human emotion labels, which we convert to our schema using the authors' official mapping.

(https://github.com/google-research/google-research/tree/master/goemotions)

- ISEAR contributes longer, self-reported emotional narratives, offering clear, unambiguous examples of each emotion.

(https://www.kaggle.com/datasets/faisalsanto007/isear-dataset)



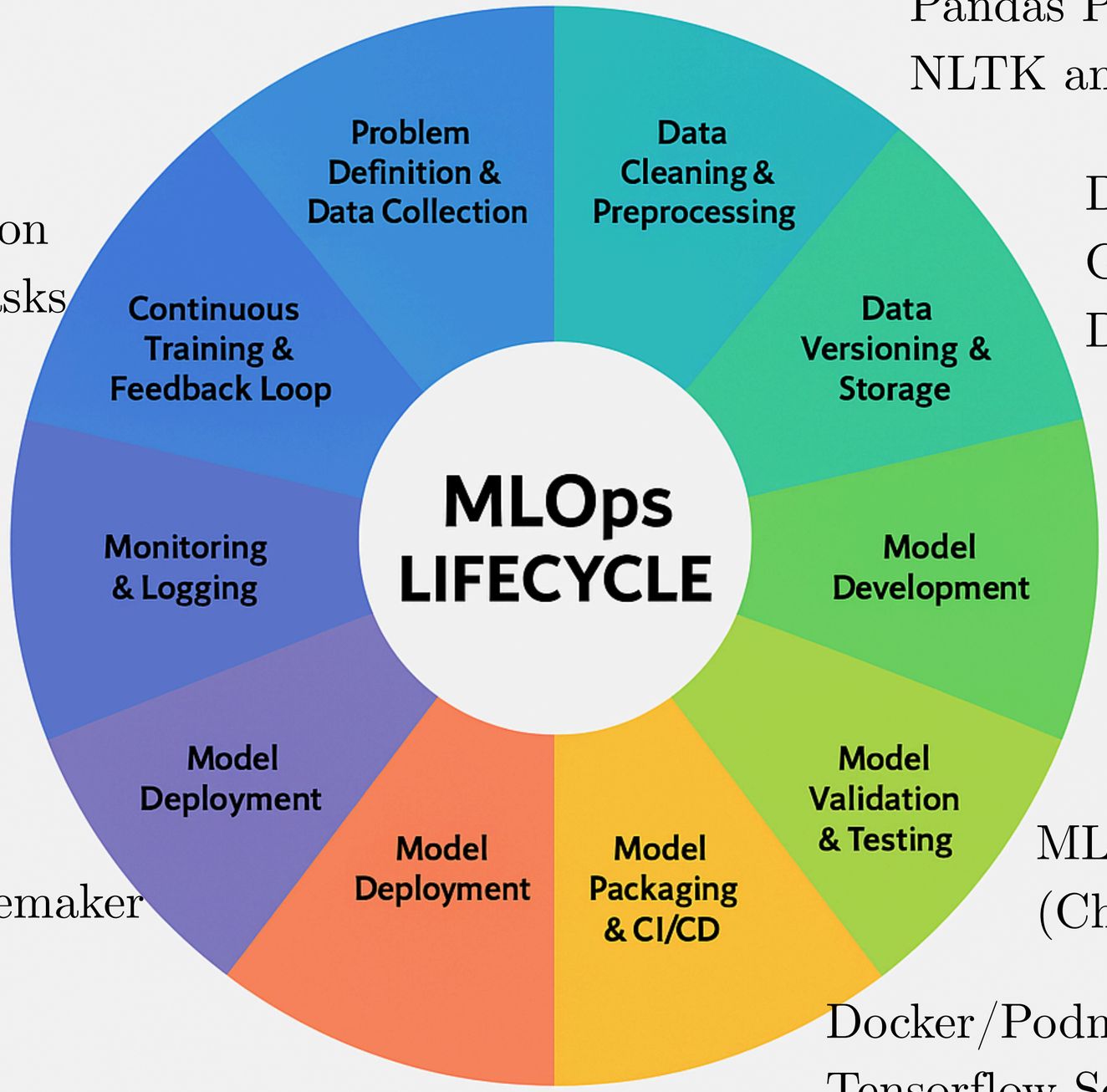
Go, change the world

Mother of MLOPS

Pandas Profiler
NLTK and submodules

DVC, Github
Google Drive
DVC Pipeline

Github Action
VS Code Tasks



Tensorflow.
keras-tuner,
RayOptuna
MLFlow

Evidently AI
Grafana
Prometheus

MLFlow registry
(Challenger & Champion)

AWS Sagemaker
Render

Docker/Podman
Tensorflow Serving

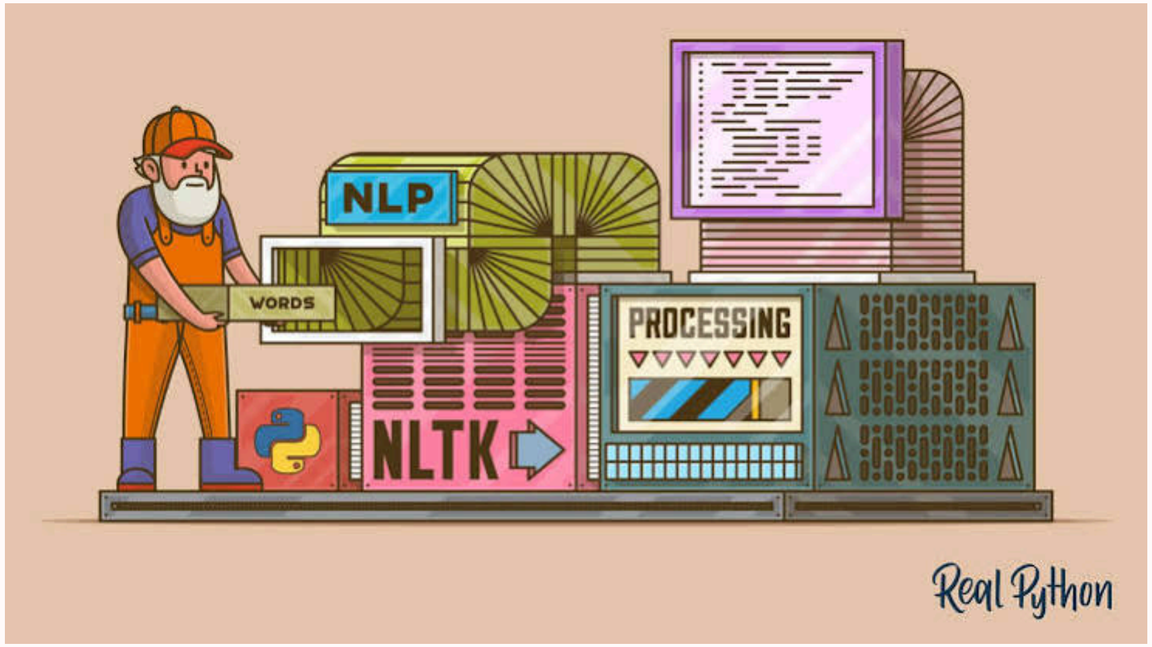
TOOLS USED

	Component	MLOPs Tools
1	Data Management	DVC, Github, Pandas Profiler
2	Experiment Tracking and Version Control	MLflow
3	Automation and Pipeline Orchestration	DVC Pipelines
4	Model Deployment & Serving	Docker,AWS SageMaker,Render,TensorflowServing
5	Monitoring & Governance and collaboration	Evidently AI,Grafana, GitHub Actions

NLTK – Text Preprocessing

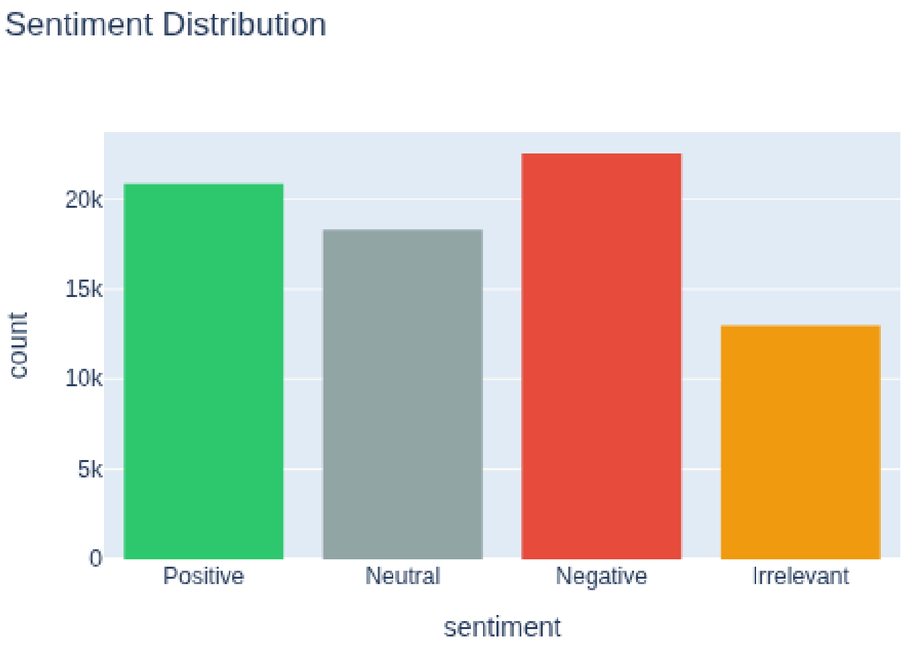
1. Always mention your packages and their versions.
2. Experiment with notebooks instead of .py
3. Use Pathlib and Ruff

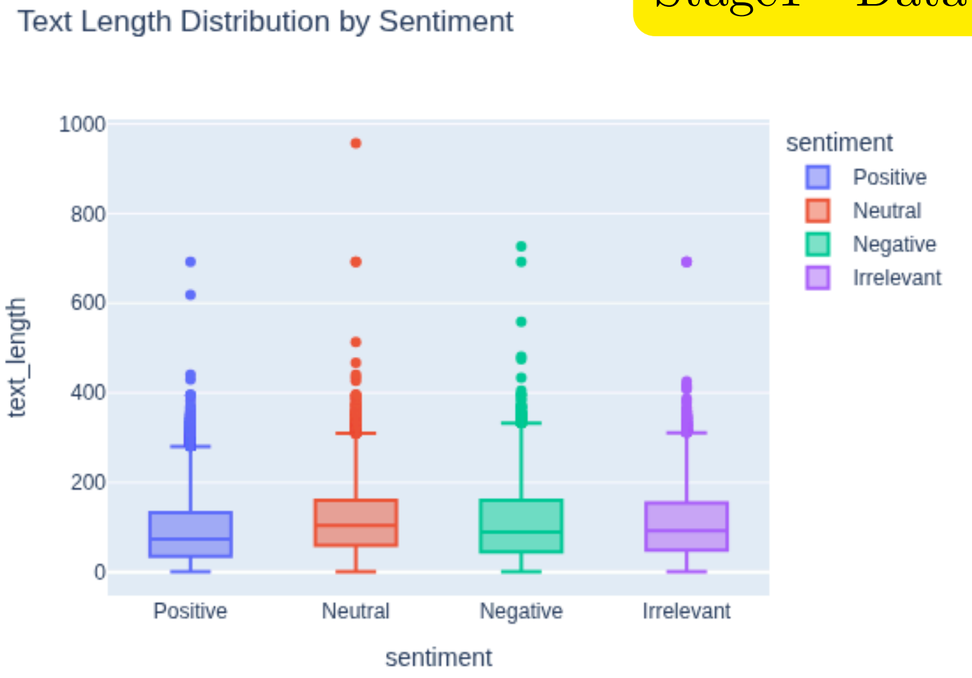
```
Environment Information
========================================
Python version      : 3.12.3
TensorFlow version  : 2.20.0
NumPy version       : 2.4.0
Pandas version      : 2.3.3
Matplotlib version  : 3.10.8
Seaborn version     : 0.13.2
NLTK version        : 3.9.2
scikit-learn version: 1.8.0
========================================
 GPU detected: /physical_device:GPU:0
```



NLTK performs tokenization, stopword removal, and lemmatization for cleaner emotion analysis input data

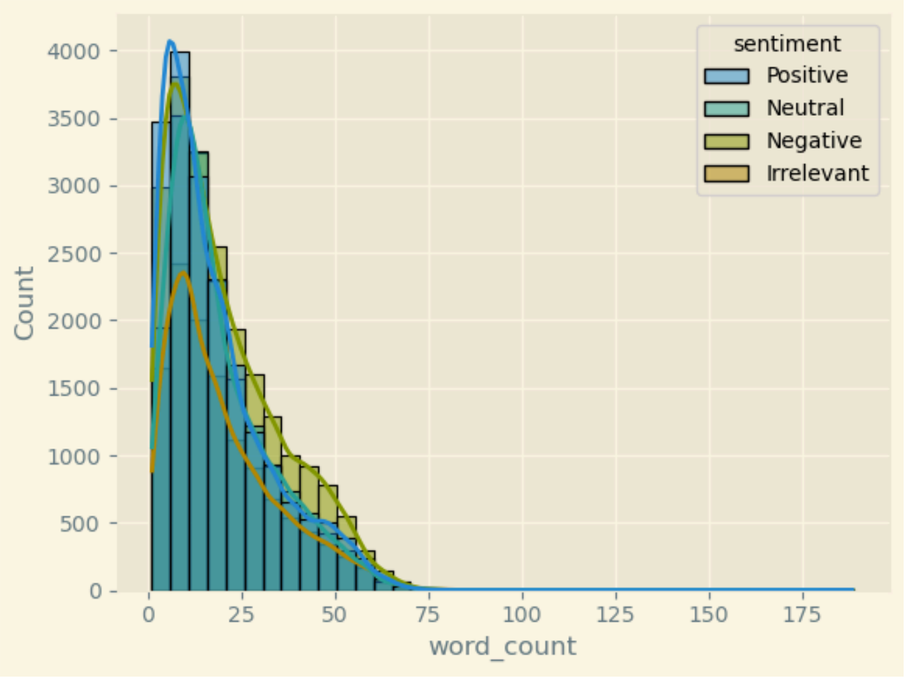
Stage1 - Data Cleaning and Processing







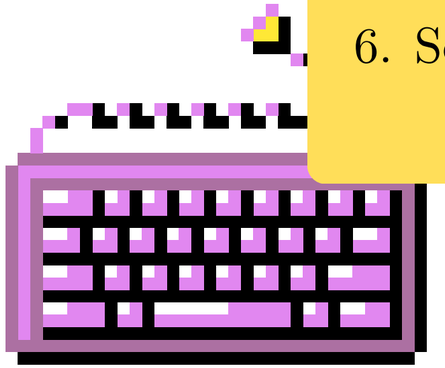


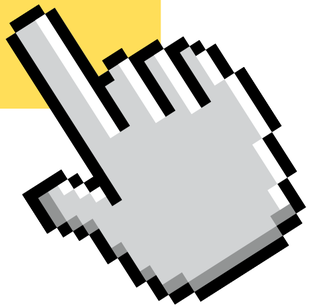


Stage1 - Data Cleaning and Processing



1. Pandas Profiler
2. NLTK/Spacy
3. Standard DL Libraries
4. Papermill (in case of python Notebooks)
5. Vanilla Pandas
6. Scikit-learn





Go, change the world

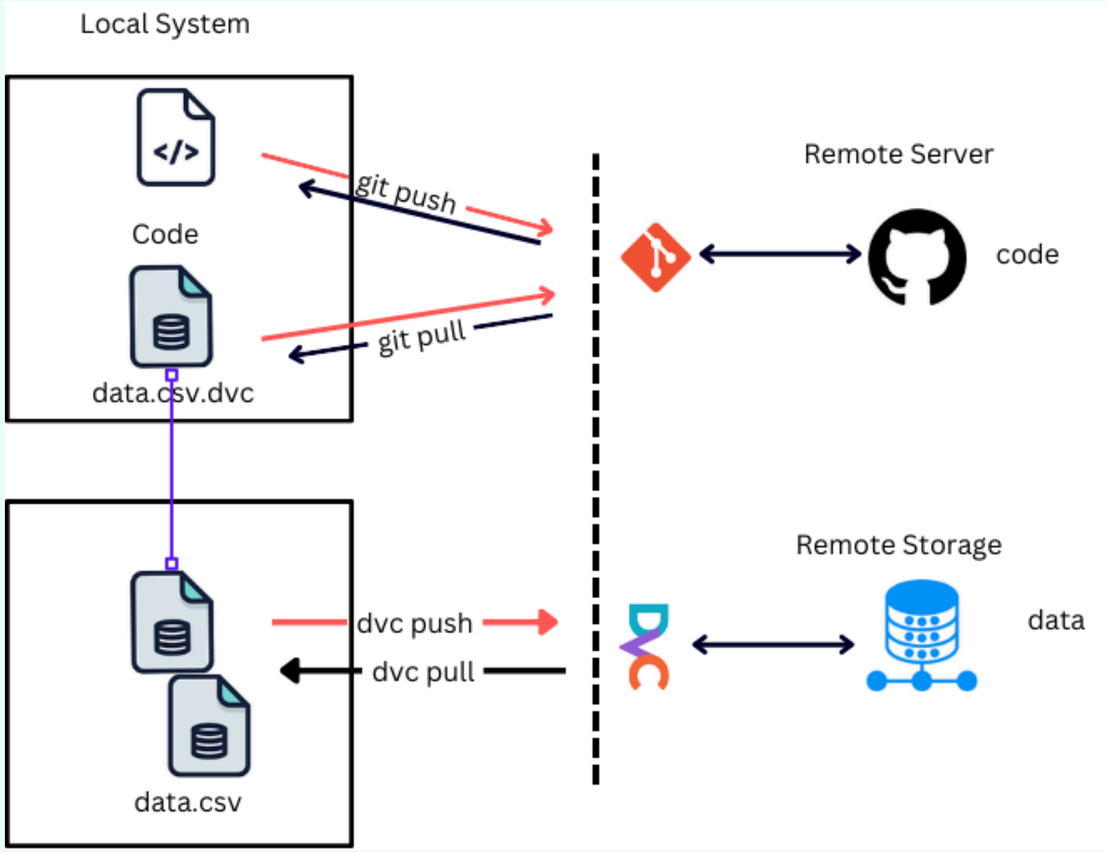
DVC – Data & Model Versioning

What is it ?

DVC tracks datasets and trained machine learning models.

Why to use ?

Machine learning depends on both code and data.



How it helps ?

When the emotion dataset or LSTM model is updated, DVC ensures the correct dataset and model versions are linked and reproducible.

DVC – Data & Model Versioning

Experiment	Created	params.yaml								
		dataset	mlflow		model	train				
		name	…riment_name	tracking_uri	architecture	batch_size	epochs	test_size	perform	…g_
workspace		Twitter	1-RNN	http://lo…	rnn	400	5	0.4	true	
main + −										
main EAI pending	07:07 PM Dec 29, 2025	Twitter	1-RNN	http://lo…	rnn	400	5	0.4	true	
8a4e602 P&G pending	05:14 PM Dec 29, 2025	Twitter	1-GRU	http://lo…	gru	400	5	0.4	true	
c6ee3de P&G pending	04:55 PM Dec 29, 2025	Twitter	1-GRU	http://lo…	gru	400	5	0.4	true	
b6b5b4c Changing some imports…	02:09 PM Dec 29, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
e579be0 Should Integrate Evid…	03:20 PM Dec 27, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
c78a81d Should add lstm anf g…	03:16 PM Dec 2, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
bb6520c Lil Bug in dockerize	02:50 PM Dec 2, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
bd9f151 Added registry folder	06:46 PM Dec 1, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
3acbd8c Render API Pending, M…	06:45 PM Dec 1, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
cb5326c Should Integrate Rend…	04:59 PM Dec 1, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	
123917f stop tracking src/art…	04:51 PM Dec 1, 2025	Twitter	1-REG	http://lo…	rnn	400	1	0.4	true	

```yaml
stages:

  preprocess:
    cmd: python -m src.data.preprocess
    deps:
      - src/data/preprocess.py
      - src/artifacts/data/Twitter/raw/train.csv
      - src/data/slangs.txt
    params:
      - dataset
    outs:
      - src/artifacts/data/Twitter/preprocessed/train.csv

  featurize:
    cmd: python -m src.data.main
    deps:
    - src/data/main.py
    params:
    - dataset
    - featurize
    outs:
    - src/artifacts/data.h5
    - src/artifacts/tokenizer.json

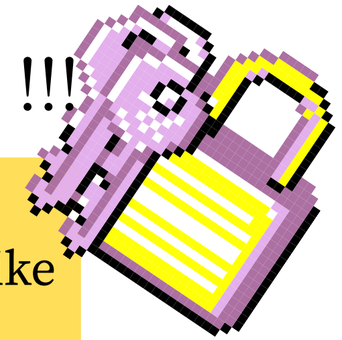
  train:
    cmd: python -m src.train.main
    deps:
    - src/train/main.py
    - src/artifacts/data.h5
    - src/artifacts/tokenizer.json
    params:
    - mlflow
    - train
    - model
    outs:
    - src/artifacts/recent:
        persist: true

  stage:
    cmd: python -m src.registry.stage
    deps:
    - src/registry/stage.py
    params:
    - train
    - model

  promote:
    cmd: python -m src.registry.promote
    deps:
    - src/registry/promote.py
    params:
    - train
    - model
```

DVC – Pipelines

Remember Versioning Data is NOT same as storing data !!!



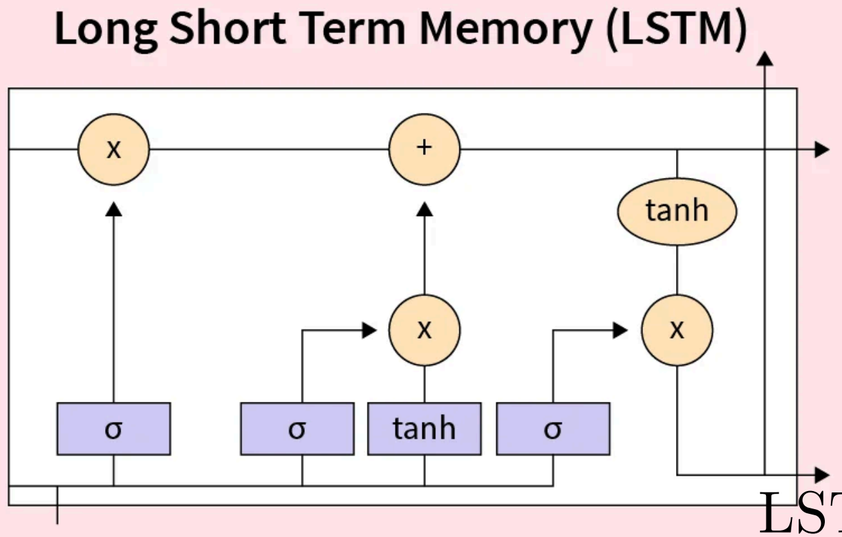
1. **lakeFS** – An open-source management platform that transforms object storage (like AWS S3, Google Cloud Storage, or Azure Blob) into a Git-like repository.
2. **Pachyderm** – A containerized, data-centric platform built on Docker and Kubernetes.
3. **Dolt** – A relational SQL database that you can clone, fork, branch, merge, push, and pull exactly like a Git repository. It versions tables rather than raw files.
4. **Delta Lake / Apache Iceberg** – Open-source storage frameworks that bring ACID transactions and time-travel version control to Spark, Presto, and large tabular data formats.
5. **Git LFS (Large File System)** – A standard Git extension that replaces large assets (like datasets or model weights) with text pointers inside Git, while storing the actual file on a remote server.



LSTM – Emotion Classification Model

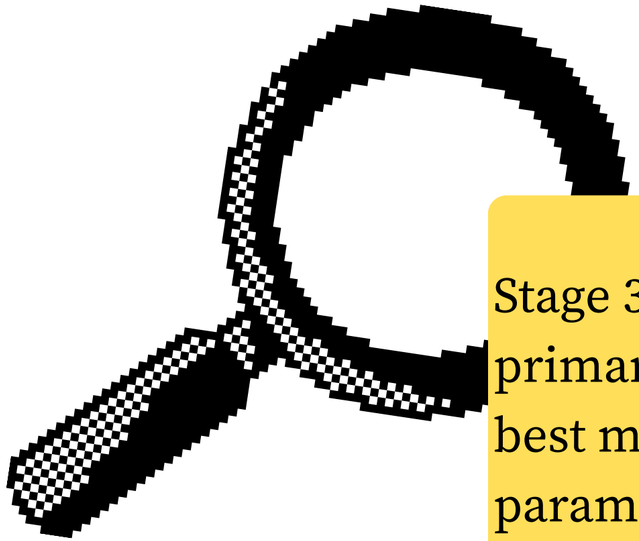
Why use it?

Text data is sequential, and the meaning of a word depends on the surrounding words.
Traditional ML models cannot capture this context effectively.



How it helps ?

LSTM captures word order and contextual information in text, enabling accurate emotion classification by understanding the emotional flow of sentences.



Stage 3.1 in Model Development primarily deals with finding the best model architecture and parameters - **Hyperparamter Tuning**

1. Tensorboard
2. keras-tuner
3. RayOptuna
4. Pytorch Lightning



MLFLOW

What is MLflow?

MLflow is an open-source MLOps platform used to manage the complete machine learning lifecycle, including experimentation, tracking, and model management.

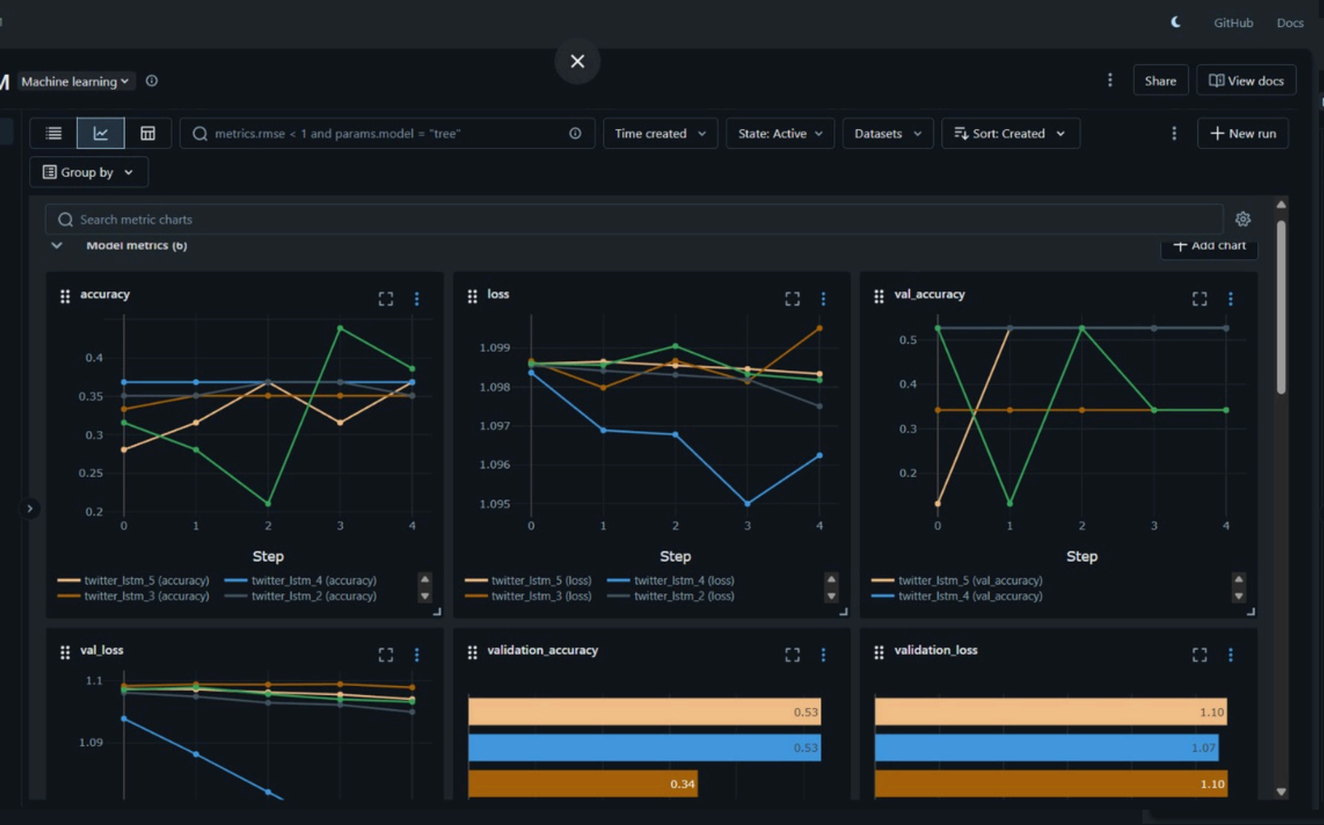
Why is MLflow used?

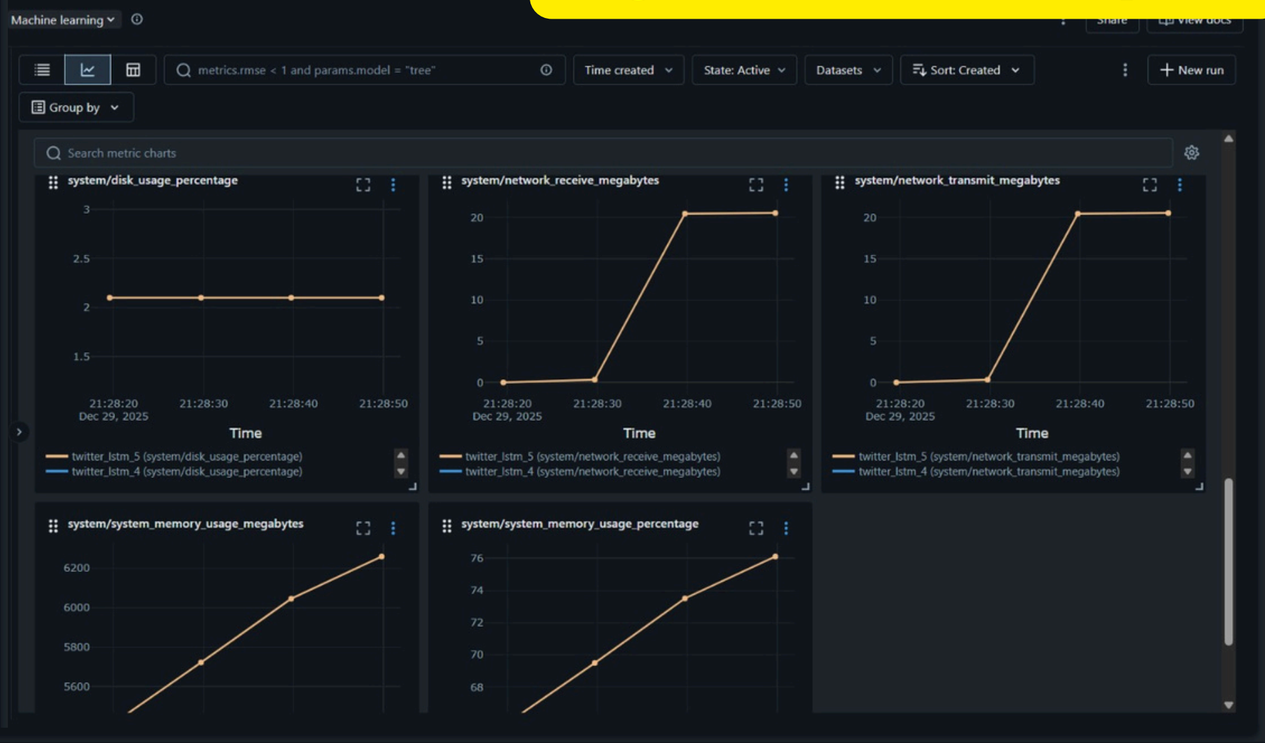
It is used to track hyperparameters, performance metrics, and model artifacts, enabling reproducibility and systematic comparison of multiple experiments.

How it helps ?

Our model involves multiple CNN configurations and training runs. MLflow ensures organized experiment tracking, version control, and reliable selection of the best-performing model, while supporting future deployment.

Stage 3.2 - Model Development





Registered Models

Share and manage machine learning models. Learn more

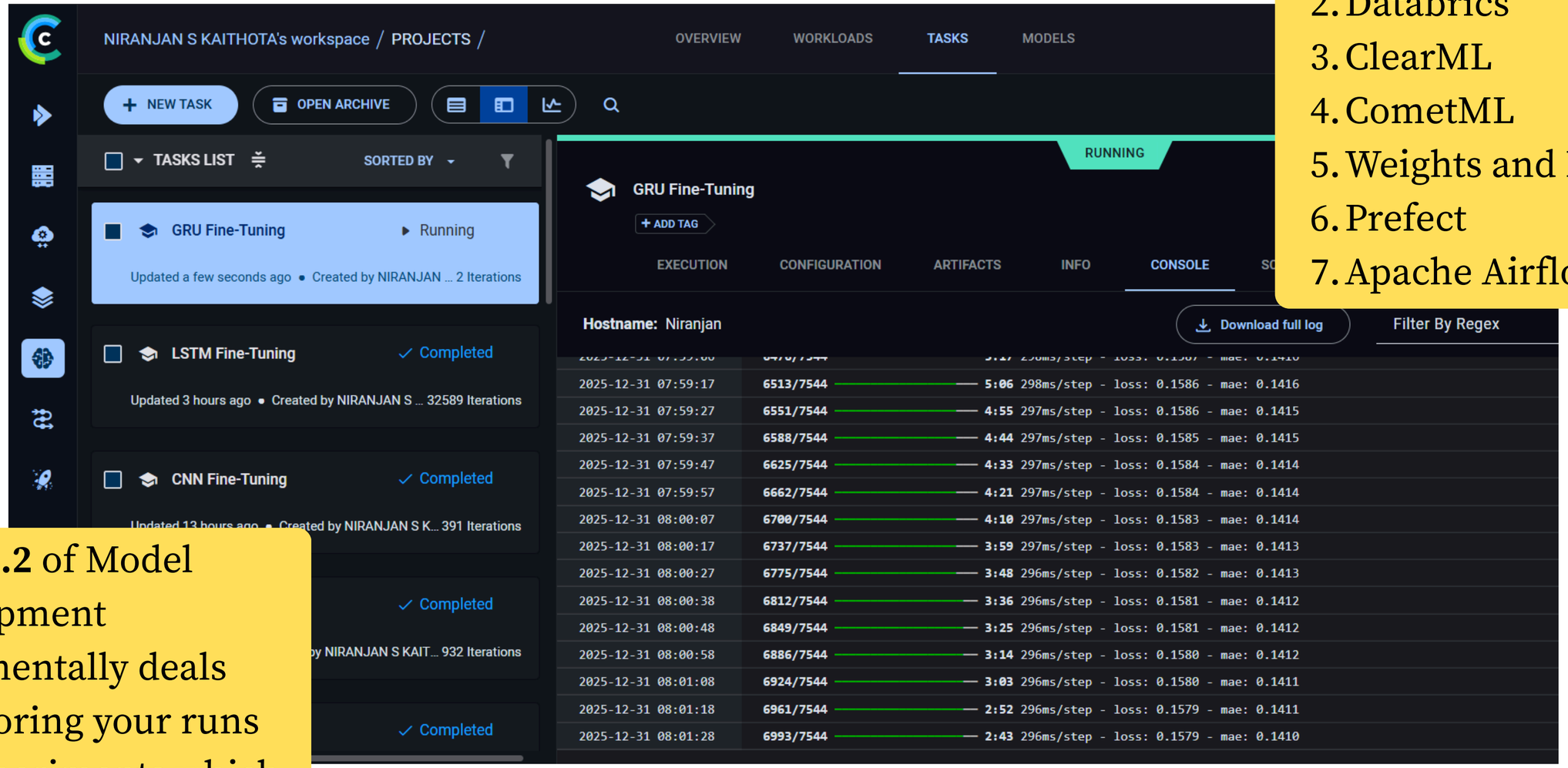
Create Model

Filter registered models by name or tags

Name	Latest version	Aliased versions	Created by	Last modified	Tags
1-GRU	Version 2	@ staged : Version 2 +1		12/29/2025, 09:59:40 ...	—
1-LSTM	Version 1	@ inprod : Version 1		12/29/2025, 08:52:12 ...	—
1-REG	Version 2	@ staged : Version 2 +1		12/29/2025, 01:24:37 ...	—
1-RNN	Version 2	@ staged : Version 2 +1		12/29/2025, 10:48:07 ...	—

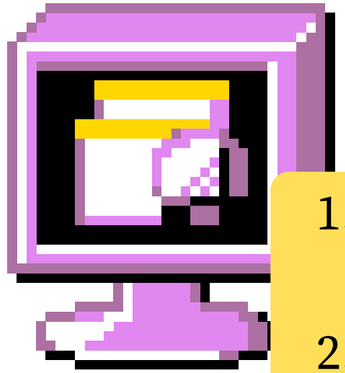
Stage 3.2 - Model Development

1. MLFlow
2. Databrics
3. ClearML
4. CometML
5. Weights and Biases
6. Prefect
7. Apache Airflow



Stage 3.2 of Model Development fundamentally deals with storing your runs and experiments which can be later **reproduced**

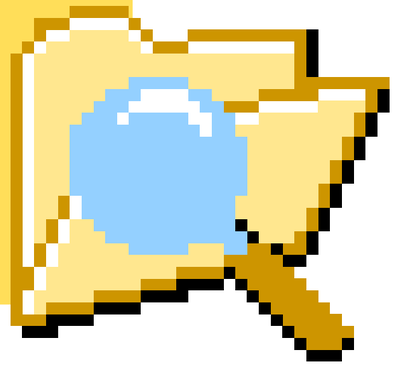
Stage 4 - Model testing and evaluation



Crucial Aspect is Model Evaluation and Testing can be based on multiple parameters

1. Based on ***Power Consumption*** and Emission
2. Based on How well you can ***Explain*** your model
3. Based on ***ethical working*** of the model
4. Based on ***DevOps*** params
5. Based on Metrics of ***production vs staged model***
6. Based on Multiple customer facing models

1. MLFlow registry (Challenger & Champion)
2. PyTest
3. Deepchecks
4. Evidently
5. Scikit-learn
6. SHAP and LIME
7. GradCAM++
8. Task Manager
9. EcoLogits
10. CodeCarbon



RV College of Engineering®
Mysore Road, RV Vidyaniketan P
Bengaluru - 560059, Karnataka,

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Stage 5 - Model packaging



ocker desktop PERSONAL

Search Ctrl+K

Stage 5 - Model packaging

Ask Gordon BETA
Containers
Images
Volumes
Builds

Models BETA
MCP Toolkit BETA

Docker Hub
Docker Scout

Extensions

Builds / SentimentAnalysis

SentimentAnalysis

uj5qb21l1ddfmp8zosv62ydgb

Status	Duration	Builder	Aut
Completed	5m 47s	desktop-linux	N/

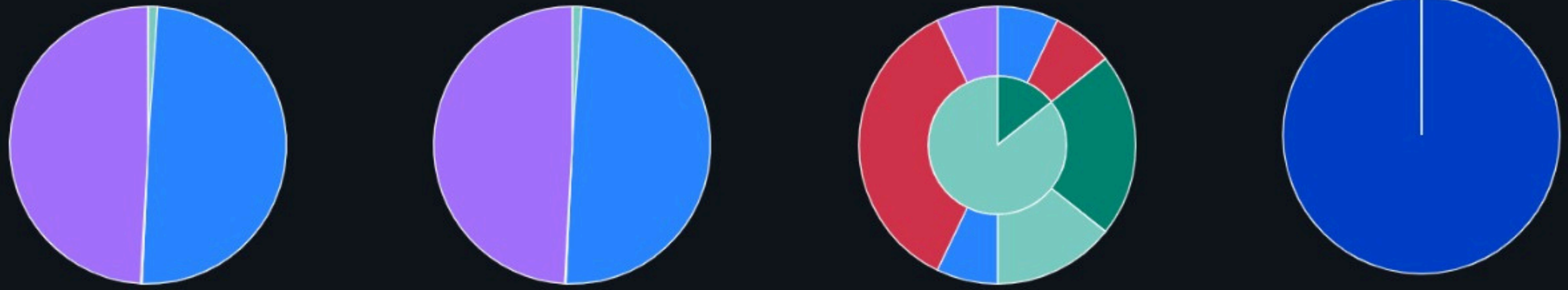
Info Source Logs History

Remote source location https://github.com/MohithTP/SentimentAnalysis.git

Revision d5cf4b2

Build timing

Real time	Accumulated time	Cache usage	Parallel execution
5m 47.3s	5m 47.3s	2/14	



Local file transfers Image pulls Executions File operations Result exports Idle

Build start time	12/29/2025, 10:27:58 PM	Cached steps	
Build end time	12/29/2025, 10:33:46 PM	Non-cached steps	

Render – Cloud Deployment and Model Hosting

What is it ?

Render is a cloud platform used to deploy machine learning applications.

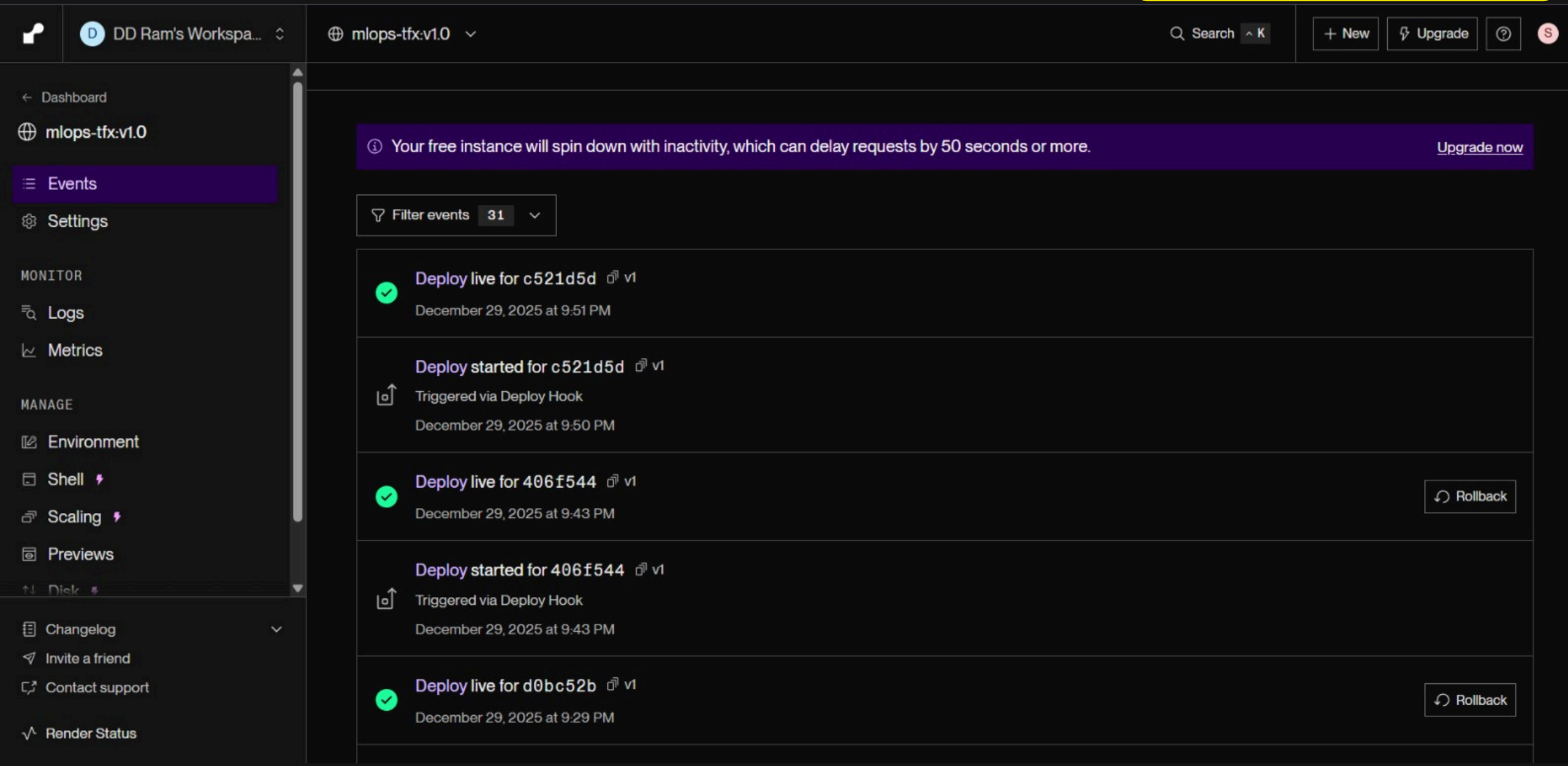
Why use it?

Motion analysis models must be accessible as a live service.

How it helps ?

Render deploys the emotion analysis model as an API, allowing real-time emotion prediction without running the model locally.

Stage 5 - Model Deployment



Stage 5 - Model Deployment

Render

01:02:04 INCOMING HTTP REQUEST DETECTED ...

01:02:07 SERVICE WAKING UP ...



01:02:11 ALLOCATING COMPUTE RESOURCES ...

01:02:14 PREPARING INSTANCE FOR INITIALIZATION ...

01:02:18 STARTING THE INSTANCE ...

01:02:24 ENVIRONMENT VARIABLES INJECTED ...

01:02:26 FINALIZING STARTUP ...

01:02:28 OPTIMIZING DEPLOYMENT ...

01:02:30 STEADY HANDS. CLEAN LOGS. YOUR APP IS ALMOST LIVE ...

START BUILDING ON RENDER TODAY →

APPLICATION LOADING

AWS – Cloud Deployment and Model Hosting

Elastic Container Registry (ECR):

- Purpose: Serving as the private repository for our versioned Docker images.
- Workflow: Automated CI/CD (GitHub Actions) builds the environment and pushes the containerized Bi-LSTM model to ECR.

SageMaker Real-Time Hosting:

- Endpoint: A fully-managed HTTPS endpoint that serves predictions in milliseconds.
- Compute: Hosted on an ml.m5.large instance, providing the optimized CPU resources needed for TensorFlow inference.
- Life-cycle: Automated deployment using the SageMaker Python SDK to create model configurations and production variants.

AWS – Cloud Deployment and Model Hosting

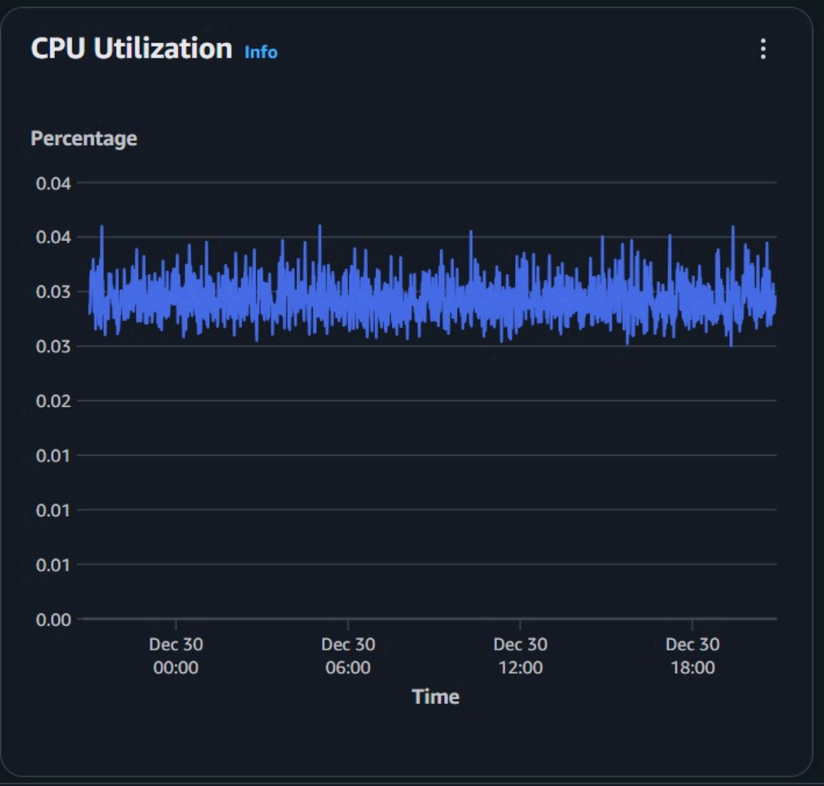
Identity & Access Management (IAM):

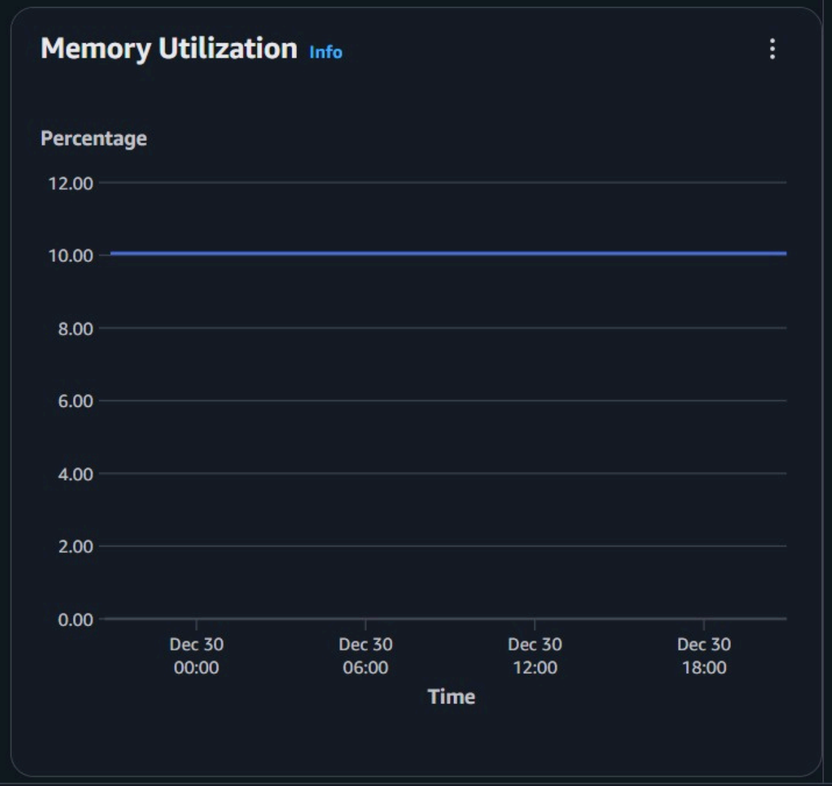
- Security: Utilizing SageMaker Execution Roles to grant "Least Privilege" access to cloud resources.
- Authentication: Local Proxy signs requests using AWS Signature Version 4 (SigV4) to ensure only authorized clients can access the model.

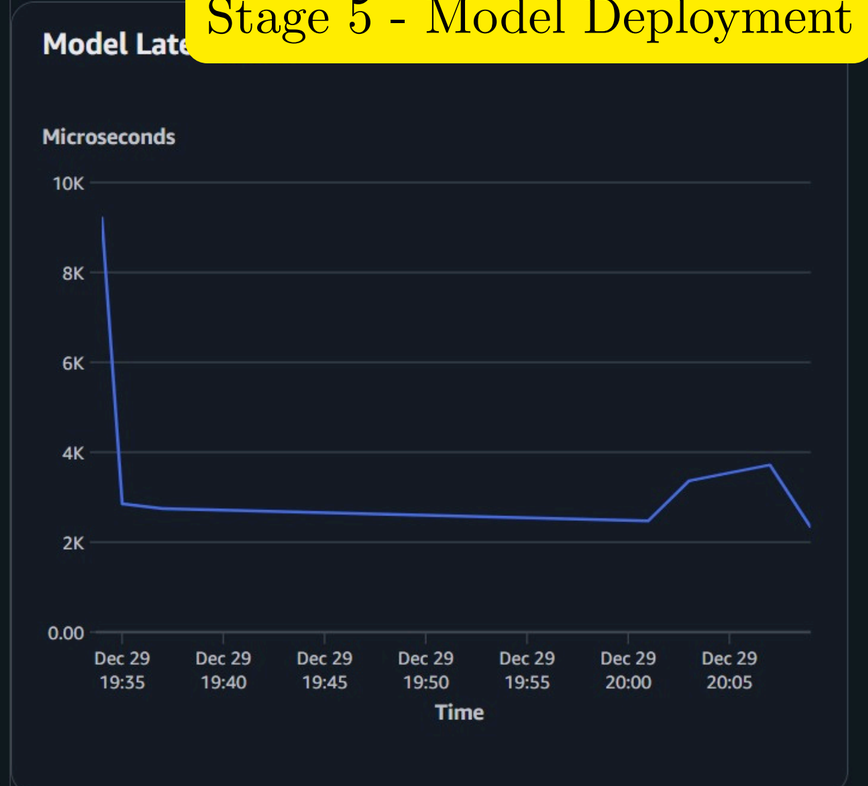
Hybrid Cloud Architecture:

- The model "lives" in the cloud, but the Local Proxy acts as a secure bridge, allowing the Chrome Extension to perform complex sentiment analysis without exposing sensitive AWS credentials.

Stage 5 - Model Deployment







Endpoints

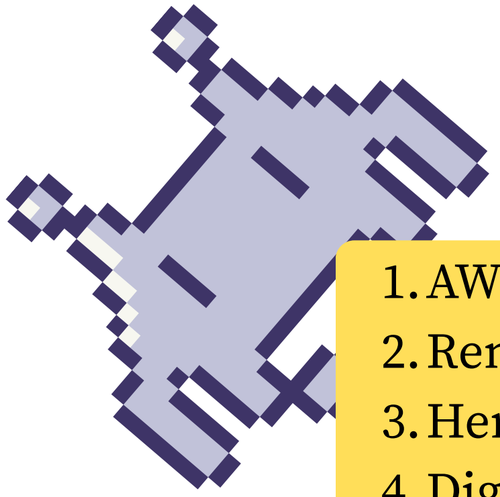
Update endpoint | Actions | Create endpoint

Search endpoints

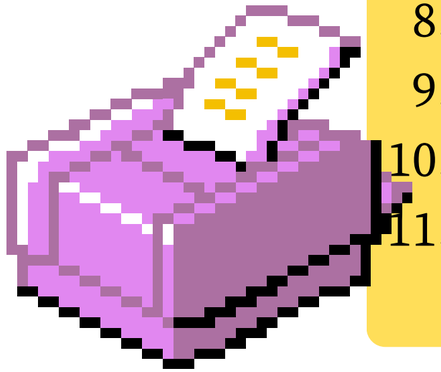
1

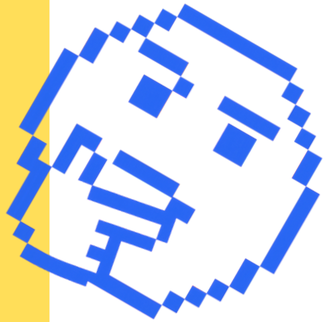
Name	ARN	Creation time	Status	Last updated
sentiment-analysis-endpoint	arn:aws:sagemaker:ap-south-1:409749468833:endpoint/sentiment-analysis-endpoint	12/29/2025, 10:20:51 AM	InService	12/29/2025, 10:23:30 AM
sentiment-analysis-v2-1767027467	arn:aws:sagemaker:ap-south-1:409749468833:endpoint/sentiment-analysis-v2-1767027467	12/29/2025, 10:44:18 PM	InService	12/29/2025, 10:47:22 PM

Stage 5 - Model Deployment



1. AWS Sagemaker
2. Render
3. Heroku
4. Digital Ocean
5. Github Pages
6. Cloudflare Pages
7. Vercel
8. E2E Networks
9. Microsoft Azure
10. Modal
11. Hugging face Inference





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Evidently AI – Data Drift Detection

What is it ?

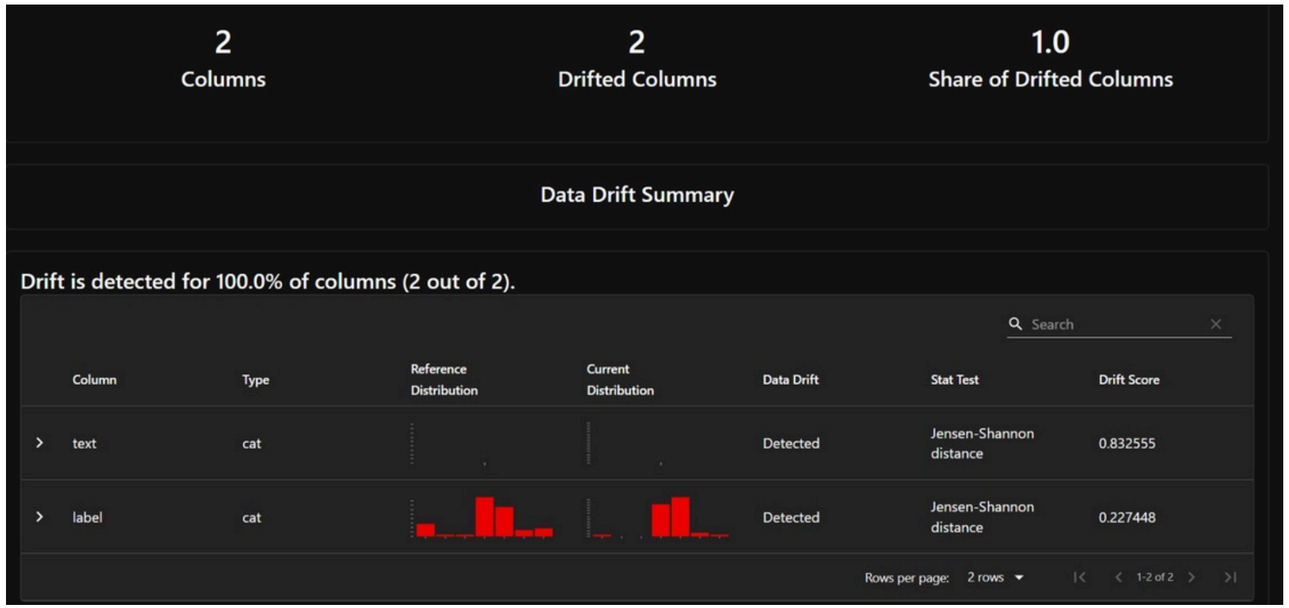
Evidently AI detects changes in data distribution over time.

Why use it?

Language and emotional expressions evolve.

How it helps ?

- Detects data drift in incoming text
- Alerts when retraining is required
- Maintains model accuracy over time



Stage 6 - Monitoring

Evidently AI – Data Drift Detection

Comparing Train Data *and* Live Logs

Overview Variables Missing values Sample Duplicate row

Select Columns

text

['Text', 'Text']

	Train Data	Live Logs
Distinct	68193	257
Distinct (%)	30.7%	30.9%
Missing	0	0
Missing (%)	0.0%	0.0%
Memory size	1.7 MiB	6.6 KiB



More details

First rows Last rows

	Train Data	Live Logs
0	id have responded if i were going	Kisko pata tha ki Kapil aaj krushna,Bharti, Simona aur Archana ko kaam dega....
1	some shameless plugging for the best rangers forum on earth	42:35 Sumona's expressions clearly tells that it was improvised
2	soooo high	Kapils struggle is not easy he cleared a lot of seasons of comedy circus love u bhai
3	both of you	38:43 Kapil spillin' some facts there!
4	as much as i love to be hopeful i reckon the chances are minimal p im never gonna g	Who's watching in 2024
5	test test from the lg env2	No one knew that Archana, Krishna, Sudesh, Bharti will work under Kapil Sharma af
6	hes just not that into you	These episodes were far better than today youtube comedians.
7	oh marly im so sorry i hope you find her soon 3 3	100000 times, Still Watching in 2025
8	is cleaning the house for her family who is comming later today	13:33 so unexpected but cute
9	gotta restart my computer i thought win7 was supposed to put an end to the constar	Sumona ka kachara kar diya

Overview Categories Words Characters

Length

	Train Data	Live Logs
Max length	8	8
Median length	7	7
Mean length	5.1722861	5.3742479
Min length	3	3

Characters and Unicode

	Train Data	Live Logs
Total characters	1149913	4466
Distinct characters	16	16
Distinct categories	1	1
Distinct scripts	1	1
Distinct blocks	1	1

The Unicode Standard assigns character

Unique

	Train Data	Live Logs
Unique	0	0
Unique (%)	0.0%	0.0%

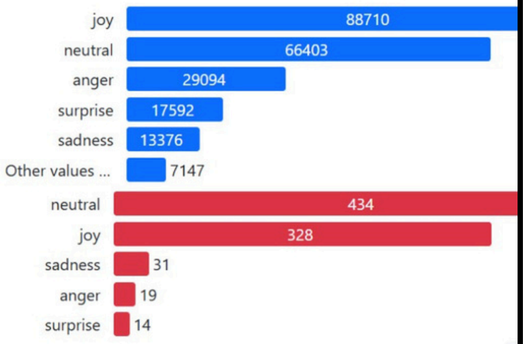
Sample

	Train Data	Live Logs
1st row	neutral	neutral
2nd row	neutral	neutral
3rd row	neutral	joy
4th row	neutral	neutral
5th row	neutral	neutral

label

Categorical

	Train Data	Live Logs
Distinct	7	6
Distinct (%)	< 0.1%	0.7%
Missing	0	0
Missing (%)	0.0%	0.0%
Memory size	1.7 MiB	6.6 KiB



Grafana & Prometheus – System & Performance Monitoring

What is it ?

Grafana is an open-source monitoring and visualization tool used to track system and application performance metrics in real time.

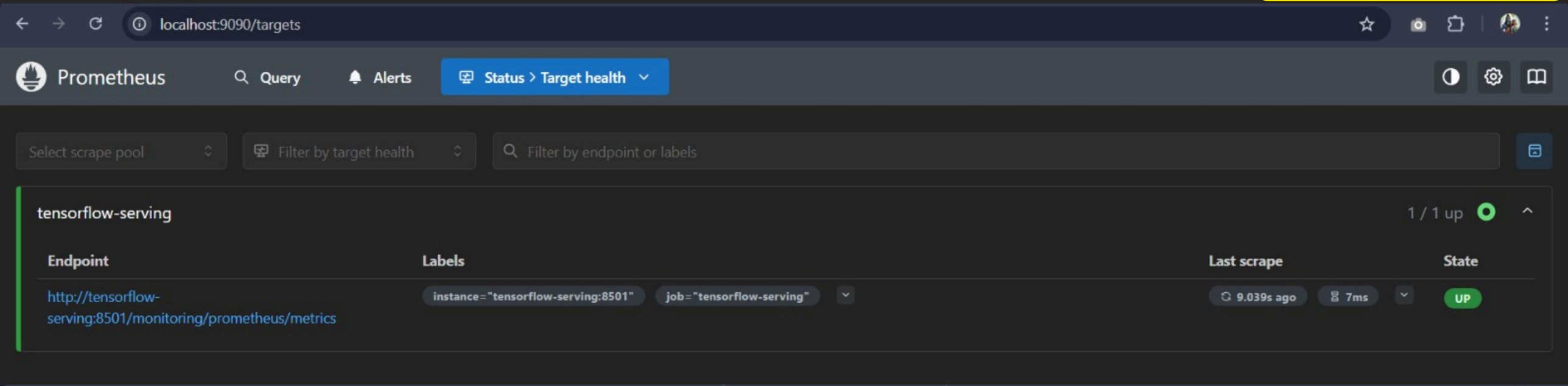
Why use it?

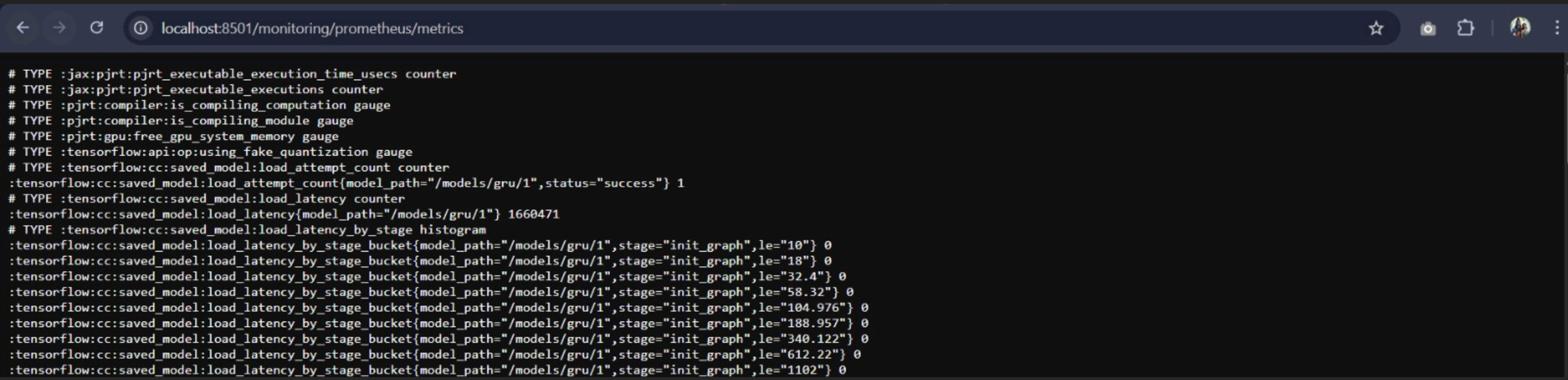
To monitor the performance of the deployed emotion analysis model.

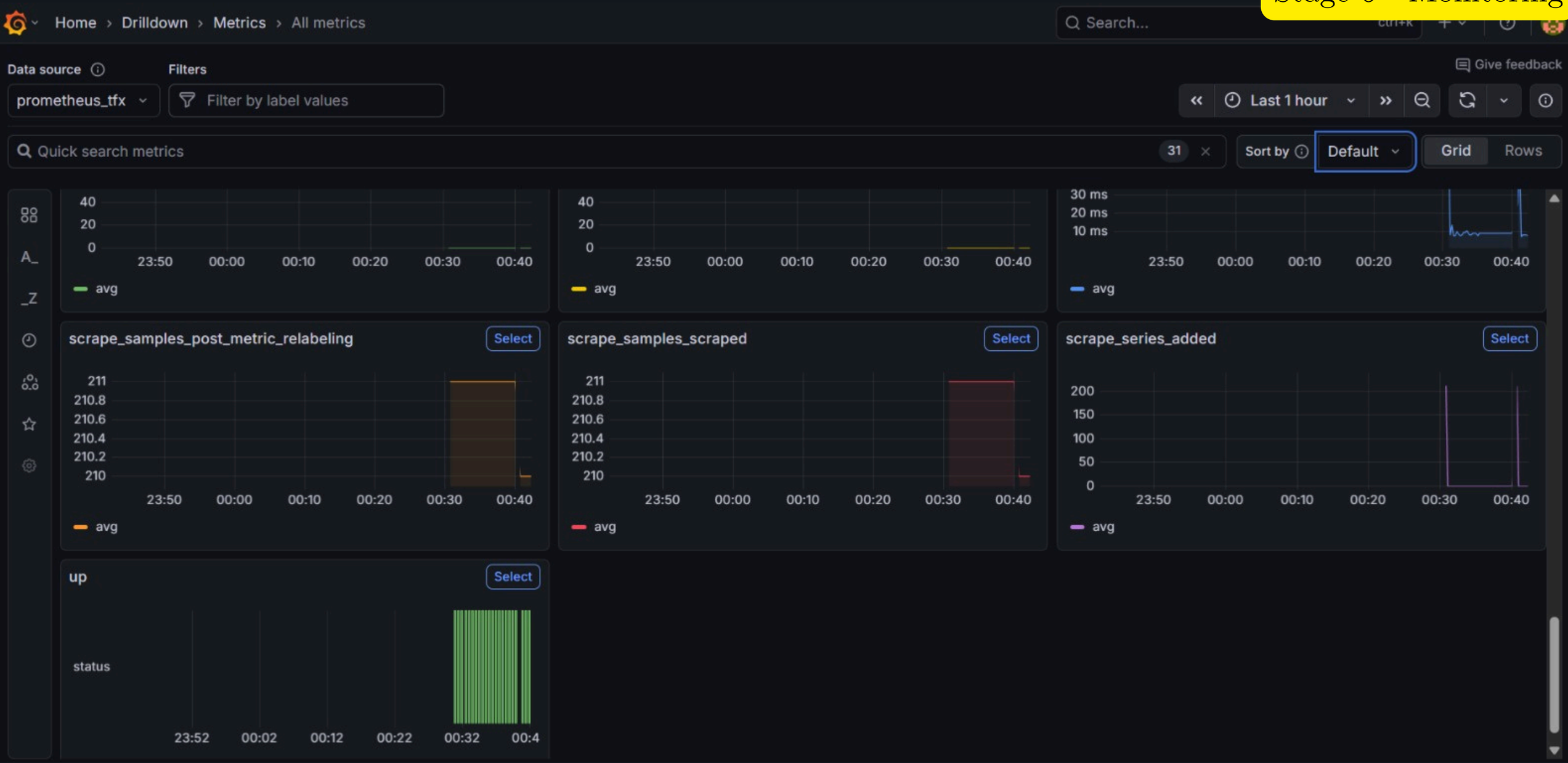
How it helps ?

Monitors CPU and memory usage
Tracks system performance during predictions

Stage 6 - Monitoring






Home › Drilldown › Metrics › All metrics
Search...
Data source
prometheus_tfx
Filters
Filter by label values
Give feedback
Last 1 hour
Quick search metrics
31
Sort by
Default
Grid
Rows
40
20
0
23:50
00:00
00:10
00:20
00:30
00:40
avg
30 ms
20 ms
10 ms
scrape_samples_post_metric_relabeling
Select
211
210.8
210.6
210.4
210.2
210
scrape_samples_scraped
scrape_series_added
200
150
100
50
0
up
status
23:52
00:02
00:12
00:22
00:32
00:4



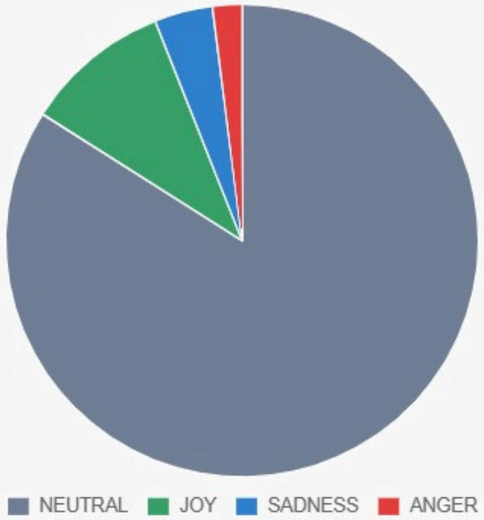
1. Evidently AI
2. AI Fairness 360
3. SHAP and LIME
4. GardCAM++
5. Prometheus and Grafana
6. Datadog
7. Newrelic
8. Tensorflow Serving
9. Custom Websocket endpoints

Post-Deploy Results

YT Sentiment AI

Analyzes visible comments

Analyze Comments



Dominant Emotion:	NEUTRAL
Positive:	10%
Sample Size:	50

First rows | Last rows

Train Data

0	id have responded if i were going
1	some shameless plugging for the best rangers forum on earth
2	soooo high
3	both of you
4	as much as i love to be hopeful i reckon the chances are minimal p im never gonna g
5	test test from the lg env2
6	hes just not that into you
7	oh marly im so sorry i hope you find her soon 3 3
8	is cleaning the house for her family who is comming later today
9	gotta restart my computer i thought win7 was supposed to put an end to the constar

Live Logs

0	Kisko pata tha ki Kapil aaj krushna,Bharti, Simona aur Archana ko kaam dega....
1	42:35 Sumona's expressions clearly tells that it was improvised
2	Kapils struggle is not easy he cleared a lot of seasons of comedy circus love u bhai
3	38:43 Kapil spillin' some facts there!
4	Who's watching in 2024
5	No one knew that Archana, Krishna, Sudesh, Bharti will work under Kapil Sharma af
6	These episodes were far better than today youtube comedians.
7	100000 times, Still Watching in 2025
8	13:33 so unexpected but cute
9	Sumona ka kachara kar diya

Stage 7- Feedback Loop

Salman के पापा ने Kapil के

www.youtube.com/watch

LeetCode YouTube SAP Github Gmail DSA Neural Network Canva

YouTube IN

kapil sharma show

@bindas637 6 months ago
Salman laughing hilarious 🤣 🤣 14:04
24 Reply

@MrArshKhan3 3 weeks ago
😂😂😂😂😂😂 Salman Khan 😍😍😍😍😍
Reply

@beautifulLife143 6 months ago
1000 barbhi dekhLu Mann nahi Bharta.😂😂😂😂😅😅😅
32 Reply

@pathan7586 8 months ago
कितनी भी बार देख लो बोर नहीं होता 😊
61 Reply

@dineshchavhan1516 4 months ago
11:42 Bhai bhai 😂😂😂😂
4 Reply

@Top5world05 1 day ago
mind blowing legendary family❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️❤️
Reply

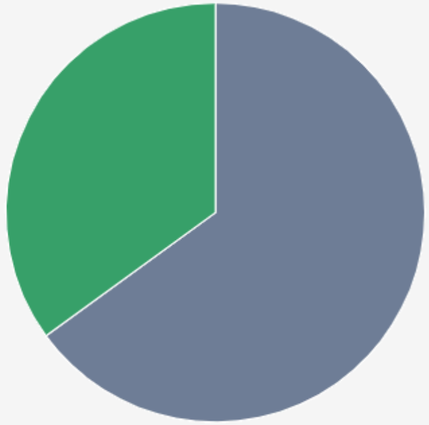
@sagarjaat3726 1 day ago
Ya vala show 😂😂😂end ha

1. Github Actions

YT Sentiment AI

Analyzes visible comments

Analyze Comments



Dominant Emotion: NEUTRAL

Positive: 35%

Sample Size: 20

HD FULL MOVIE
READY
2:25:06
Khan, Asin | Anees Bazmee, ...
T-Series
93M views • 3 years ago

शत्रुघ्न ने कपिल से शेयर की अपनी और बीवी की शादी की स्टोरी 🤣 🤣 |Th...

SriRam's Suggestions

1. Select Industry Standard Tools than the newer ones.
2. Check for version conflicts, especially with Tensorflow and Pytorch CPU Versions and GPU Versions
3. For sure check the VERSIONS !!! and maintain a proper Github Commit History
4. Learn how to merge different commits
5. Always, ALWAYS use Standard python libraries to make your code base cleaner – pydantic schemas, Pathlib file paths, Loguru (Logger), Rich for terminal printing
6. Select standard set of compatibles tools like the – Kubernates Family or MLFLow-Tensorflow Family
7. Learn to maintain proper directory structure, it is the key to any project (Look into Cookiecutter Templates if needed)
8. Always write your own documentation/journey.md / README.md file.
9. Try to run things offline rather than being dependent on some online hopeless people
10. Make sure to go step by step and make sure you dont push more than 100MB files in github, always maintain .gitignore and .dockerignore files
11. Focus more on learning the MLOPS tools in this subject rather than worrying about actual metrics of model

Department of AI and ML

